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UNIVERSITI
TEKNOLOGI
MARA

COLLEGE OF COMPUTING, INFORMATICS AND MATHEMATICS

ICT502 : DATABASE ENGINEERING

TITLE: BOUTIQUE MANAGEMENT SYSTEM

PREPARED BY:

BIL	MATRIC NO.	NAME	GROUP
1.	2024239076	ROHADATUL AKMA BINTI RADZUAN	CDCS2302C GROUP 3
2.	2024205198	INTAN SYAFIKA BINTI MOHD AZMI	
3.	2024203366	NURUN NAJWA BINTI NASROLLAH	
4.	2024420362	NUR ARIFAH NAJMIE BINTI ISHAK	
5.	2024288208	NUR AMIRA IZREENA BINTI MOHAMAD AZLAN	

PREPARED FOR :

DR. NOR DIANA BINTI AHMAD

BACHELOR OF SCIENCE COMPUTERS (WITH HONOURS) CDCS230

NAME	PROFILE PICTURE
ROHADATUL AKMA BINTI RADZUAN	
INTAN SYAFIKA BINTI MOHD AZMI	
NURUN NAJWA BINTI NASROLLAH	
NUR ARIFAH NAJMIE BINTI ISHAK	
NUR AMIRA IZREENA BINTI MOHAMAD AZLAN	

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1.0 COMPANY BACKGROUND



Figure 1.1 Logo of Inrora Boutique

Inrora Boutique is a family-friendly clothing store that offers stylish and affordable fashion for women, men, and children. Inrora Boutique brings together modern designs, good quality, and reasonable prices, making it easy for families to find everything they need in one place.

The boutique is designed to feel elegant but welcoming for all ages. Customers are invited to visit the store in person to see and try the clothes themselves. Inrora Boutique does not have an online store, as it believes in giving customers a personal and enjoyable shopping experience. To reward our loyal customers, Inrora Boutique also offers a membership program. Regular shoppers can purchase a membership to enjoy special benefits, including exclusive discounts, early access to new collections, and other exciting promotions.

At Inrora Boutique, you'll find clothing for everyday wear as well as special occasions, with styles that are both fashionable and practical. The store is proud to create a space where families can shop together and find clothes that make them feel confident and stylish.

Inrora Boutique — where fashion meets family.

2.0 CASE STUDY

2.1 PROBLEM STATEMENT

Inrrora Boutique is family-friendly clothing stores that provides fashionable yet reasonably priced clothing to a wide range of customers. Currently, the Inrrora Boutique Management System manages the core operations manually such as inventory and stock tracking, sales and invoice process and customer membership program, which causes certain problems to occur such as :

Inefficient inventory and stock management:

Manual inventory tracking causes inconsistencies between recorded and actual inventory/stock levels. Each product is physically counted by staff regularly, which is time consuming and prone to human error. This will lead to overstock for some products while other products may be suddenly out of stock and it will affect customer satisfaction and lead to potential loss of sales.

Time-consuming sales processing:

The current manual sales and invoicing process could slow down the checkout process, creating longer wait times for customers, but also increases the likelihood of calculation errors. The client experience is greatly impacted by this inefficiency during peak times.

Manual customer membership tracking:

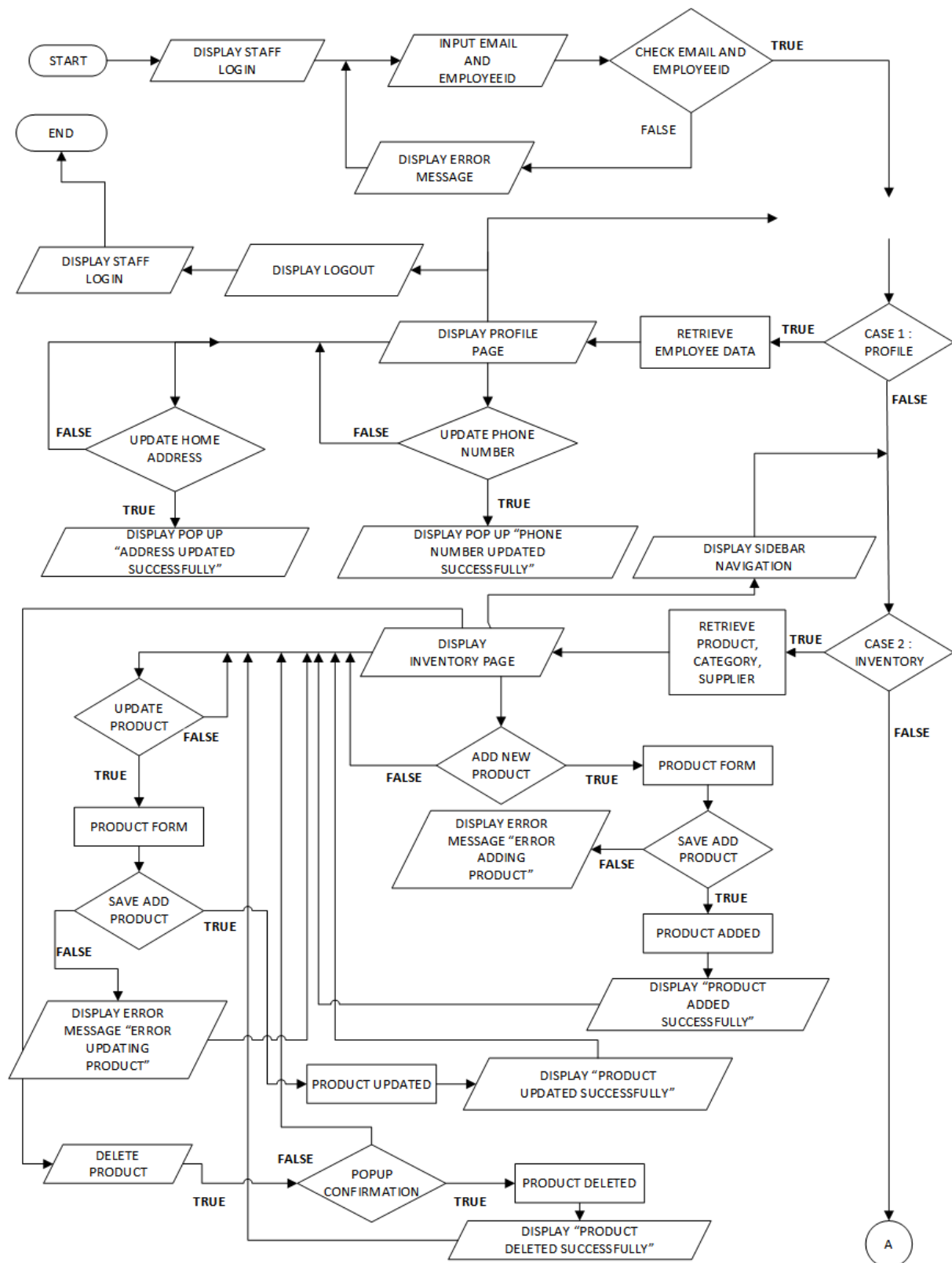
Managing the customer loyalty program manually makes it difficult to accurately track member benefits, purchase history, and eligibility for special promotions. This makes it harder to identify and reward valuable consumers and results in inconsistent discount implementation, which could compromise the membership program's efficacy.

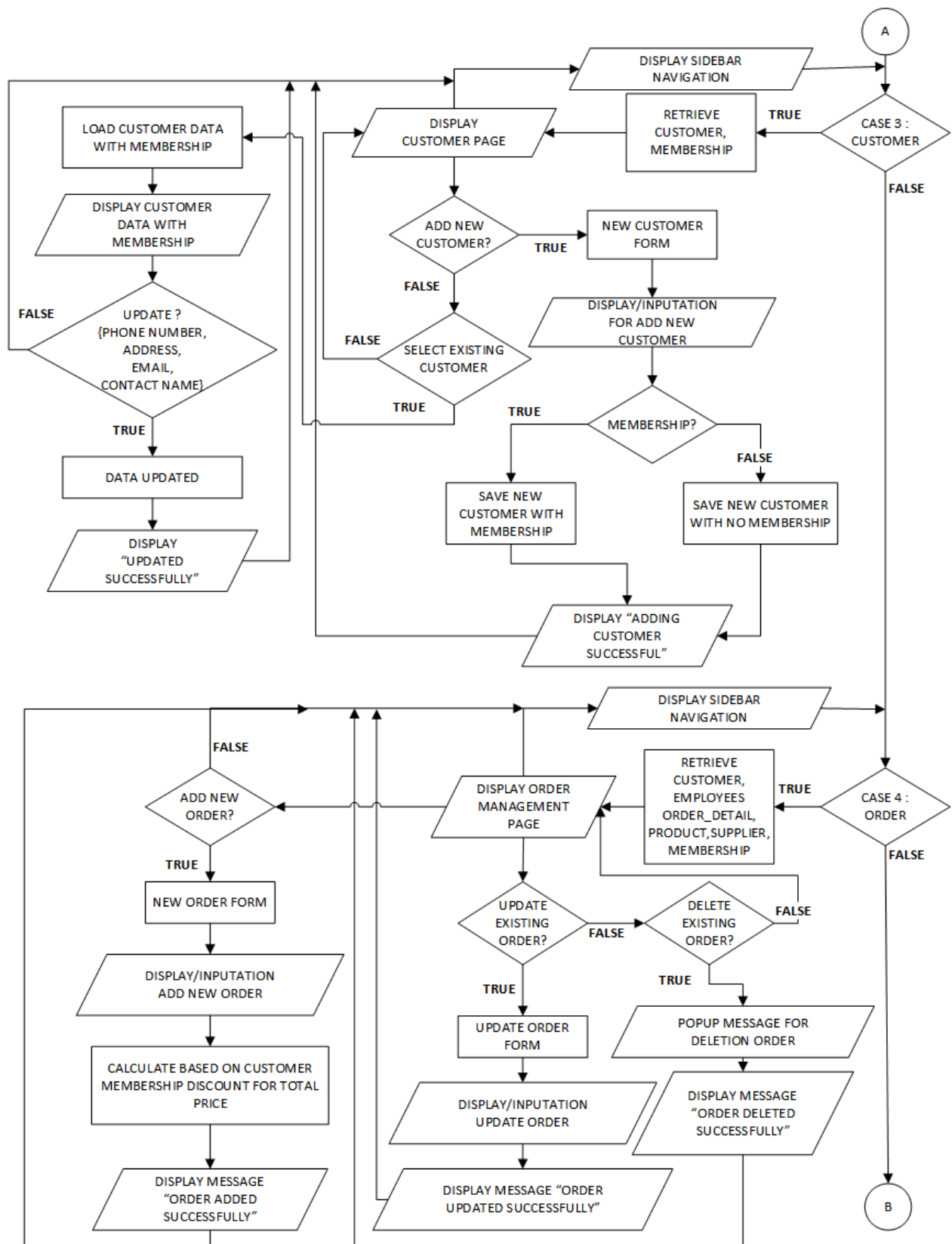
2.2 OBJECTIVE

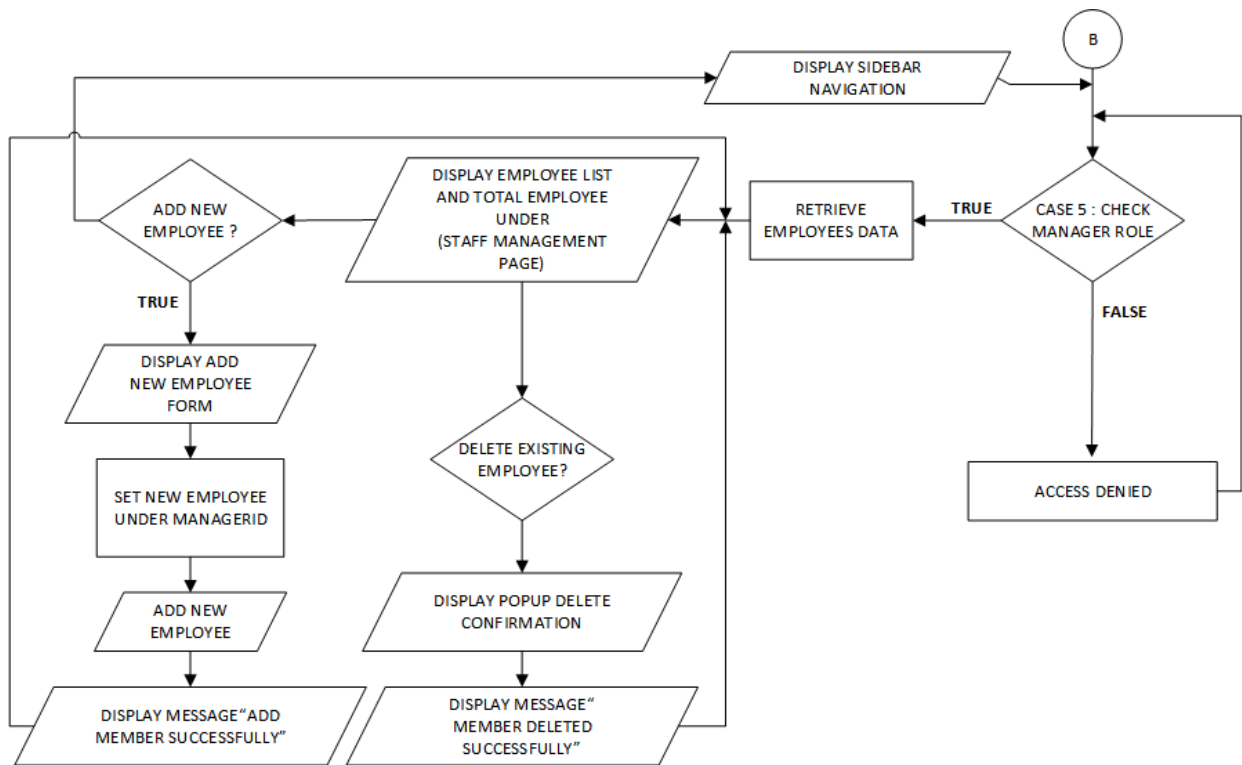
- i. To identify the requirement for the INRORA Boutique Management System
- ii. To design INRORA Boutique Management System
- iii. To develop the INRORA Boutique Management System

3.0 SYSTEM DESIGN

3.1 Flow Chart of System







3.2 10 SQL Queries

1. Retrieve customer name and their product name who placed order in September.

```
SELECT c.cust_name, o.orderdate, p.productname
FROM CUSTOMER c
JOIN ORDERS o
ON o.customerid = c.customerid
JOIN PRODUCTS p
ON p.productid = o.productid
WHERE TO_CHAR (o.ORDERDATE,'MM') =9;
```

	CUST_NAME	ORDERDATE	PRODUCTNAME
1	Mimi	14/09/2025	MuseFleck Top
2	Adi	10/09/2025	ForgeFit Joggers
3	Hana	21/09/2025	EverMode Hoodie

2. Display employee who has no manager and the order ID for orders they managed.

```
SELECT e.employeeid, e.firstname || ' ' || e.lastname AS FULLNAME,
e.emp_phonenum, e.emp_email, o.orderid
FROM EMPLOYEES e
JOIN ORDERS o
ON o.employeeid= e.employeeid
WHERE e.managerid IS NULL
```

	EMPLOYEEID	FULLNAME	EMP_PHONENUM	EMP_EMAIL	ORDERID
1	111	Marina Kassim	012 3334440	siti.k@inrora.com	6100

3. Display total orders managed by employee with ID of 115.

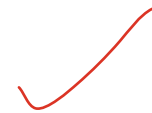
```
SELECT COUNT(orderid) AS "Orders Managed by Employee 115"
FROM ORDERS
WHERE employeeid = 115;
```



	Orders Managed by Employee 115
1	2

4. Calculate the discounted price for customers who made their orders.

```
SELECT c.customerid, c.cust_name, o.orderid, od.price*(1-m.discount) AS "PRICE
AFTER DISCOUNT"
FROM ORDER_DETAILS od
JOIN ORDERS o
ON o.orderid = od.orderid
JOIN MEMBERSHIP m
ON m.customerid = o.customerid
JOIN CUSTOMER c
ON c.customerid = m.customerid
```



	CUSTOMERID	CUST_NAME	ORDERID	PRICE AFTER DISCOUNT
1	201	Aisyah	6100	566.685
2	203	Luqman	6103	52.1475
3	204	Aliya	6104	377.38748
4	207	Farah	6107	126.0
5	208	Adi	6108	92.42999
6	209	Akmal	6109	1728.3824
7	211	Hana	6111	1158.948
8	213	Irfan	6112	3638.7651
9	214	Balqis	6113	4981.1445
10	215	Hafiz	6114	3992.5168

5. Compare length of product name and sort it in descending order.

```
SELECT productname, LENGTH(productname) as expr1
FROM PRODUCTS
ORDER BY expr1 DESC
```

	PRODUCTNAME	EXPR1
1	NeutralNest oversized Tee	25
2	Lilac Loop Cardigan	19
3	OrbitLayer Crewneck	19
4	Velvet Sway blouse	18
5	Aether Code Shirt	17
6	Sunbeam Dot Dress	17
7	SprinkleJoy Skirt	17
8	ForgeFit Joggers	16
9	IronThread Denim	16
10	TinyBloom Romper	16
11	PixelRex pajamas	16
12	EverMode Hoodie	15
13	RocketDash Tee	14
14	WinkBerry Top	13
15	MuseFleck Top	13

6. Find the total number of customers.

```
SELECT COUNT(*) AS total_customer
FROM CUSTOMER;
```

	TOTAL_CUSTOMER
1	15

7. Calculate the average price of all the products.

```
SELECT AVG(UnitPrice) AS average_price
FROM PRODUCTS;
```

	AVERAGE_PRICE
1	33.024666

8. Insert a new customer and their information details.

```
INSERT INTO CUSTOMER(customerID, cust_name, cust_PhoneNum,  
cust_Address, cust_Email, contactName)  
VALUES(216, 'Laila', '0142367821', 'No 18, Jalan Bukit Indah, 47000 Sungai Buloh,  
Selangor', 'laila@gmail.com', 'Miss Laila');
```

16	216	Laila	0142367821	No 18, Jalan Bukit Indah, 47000 Sungai Buloh, Selangor	laila@gmail.com	Miss Laila
----	-----	-------	------------	--	-----------------	------------

9. Update employee's phone number id of 208 in the customer table.

```
UPDATE CUSTOMER  
SET cust_phonenum = '0196321435'  
WHERE customerid = 208;
```

8	208	Adi	0196321435	No 40 Jalan Tok Janggut, 23000 Dungun, Terengganu	adi@gmail.com	Mr. Adi
---	-----	-----	------------	---	---------------	---------

10. Display total stock which available for each category.

```
SELECT ca.categoryName, SUM(p.UnitStock) AS totalStock  
FROM PRODUCTS p  
JOIN CATEGORY ca ON p.categoryID = ca.categoryID  
GROUP BY ca.categoryName;
```

	CATEGORYNAME	TOTALSTOCK
1	Men Cloth	3500
2	Women Cloth	3000
3	Children (girl) Cloth	5500
4	Children (boy) Cloth	3000
5	Unisex Cloth	4000

3.3 System Development Sample Screen

Customer Management

Select Customer:
Adi (ID: 208)

Load Customer + Add New Customer

Customer ID: 208
Customer Name: Adi
Membership Status: Member (Discount: 10%)

Phone Number: 017 8853421 Update

Address: No 40 Jalan Tok Janggut, 23000 Dungun, Terengganu Update

Email: adi@gmail.com Update

Contact Name: Mr. Adi Update

Figure 3.3.1 Read Customer Page

Add New Customer

Customer Name: *
Enter customer name

Phone Number:
Enter phone number

Address:
Enter address

Email:
Enter email

Contact Name:
Enter contact name

Membership Options:
☒ No Membership
☐ Add Membership

* Required fields

Add Customer Cancel

Figure 3.3.2 Insert Customer Page

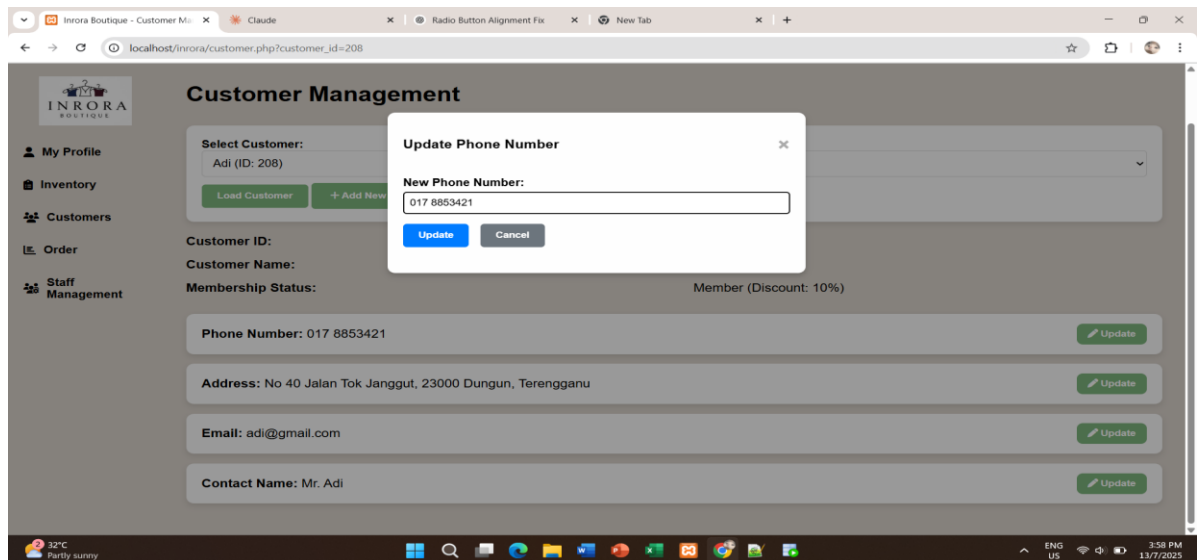


Figure 3.3.3 Update Customer Page

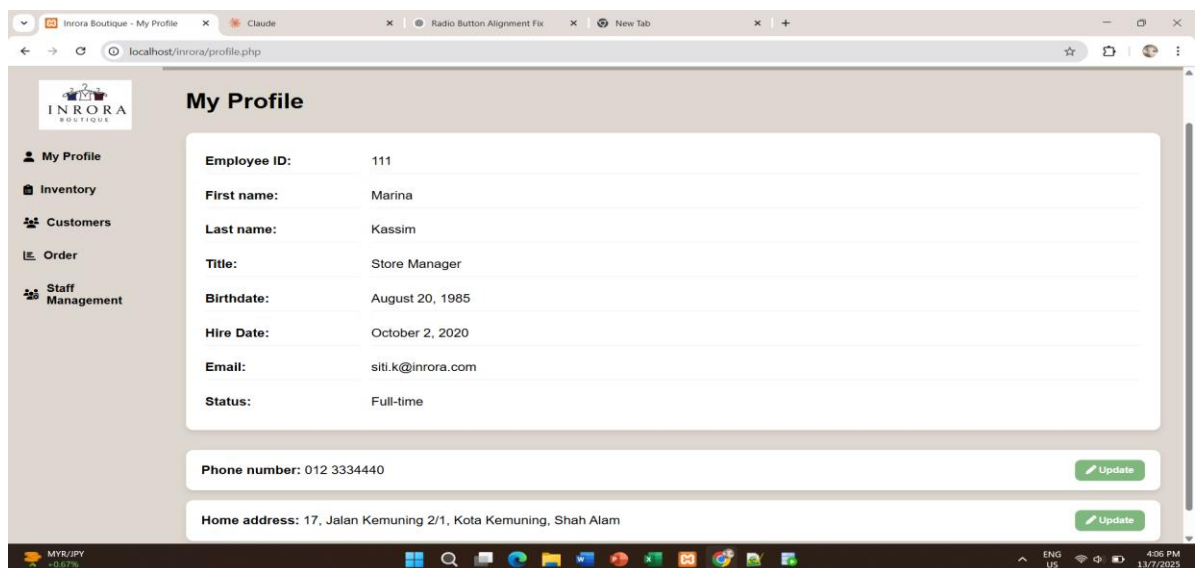


Figure 3.3.4 Read Employee Page

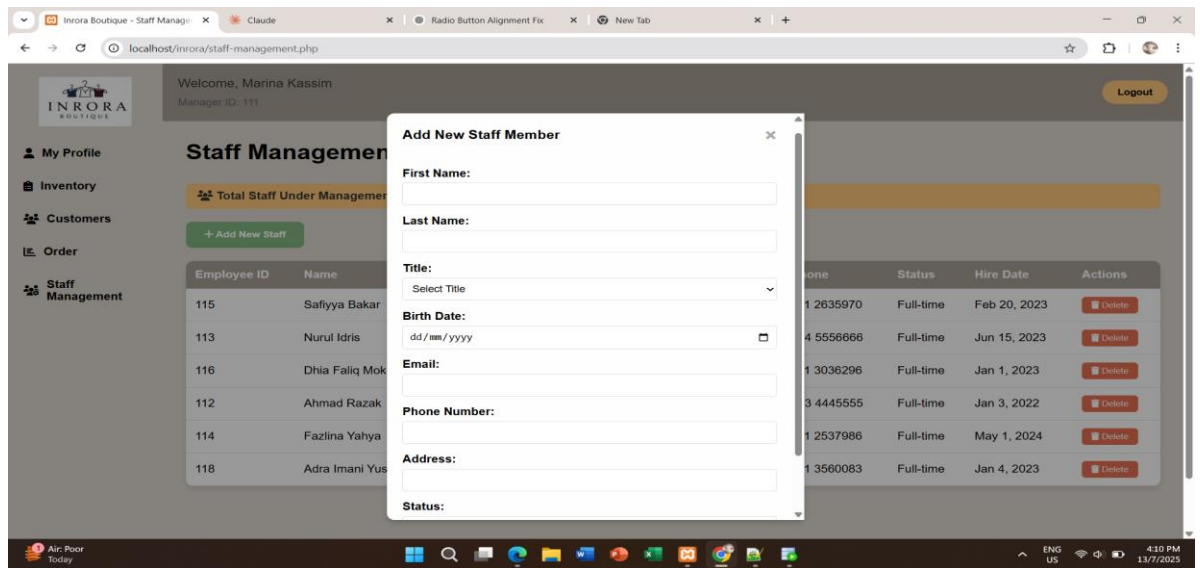


Figure 3.3.5 Insert Employee Page

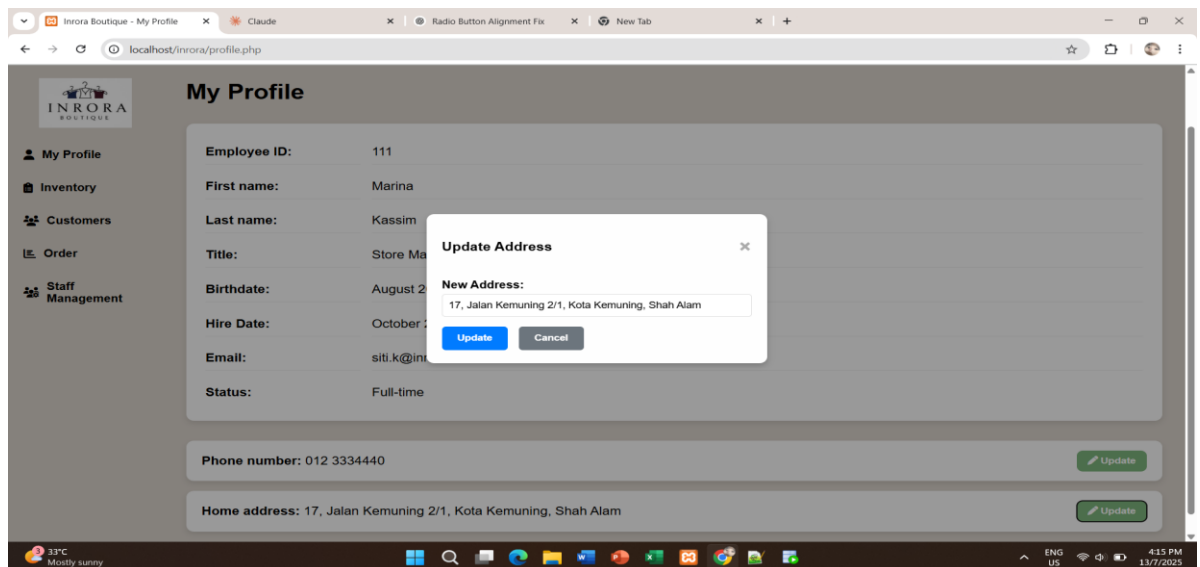


Figure 3.3.6 Update Employee Page

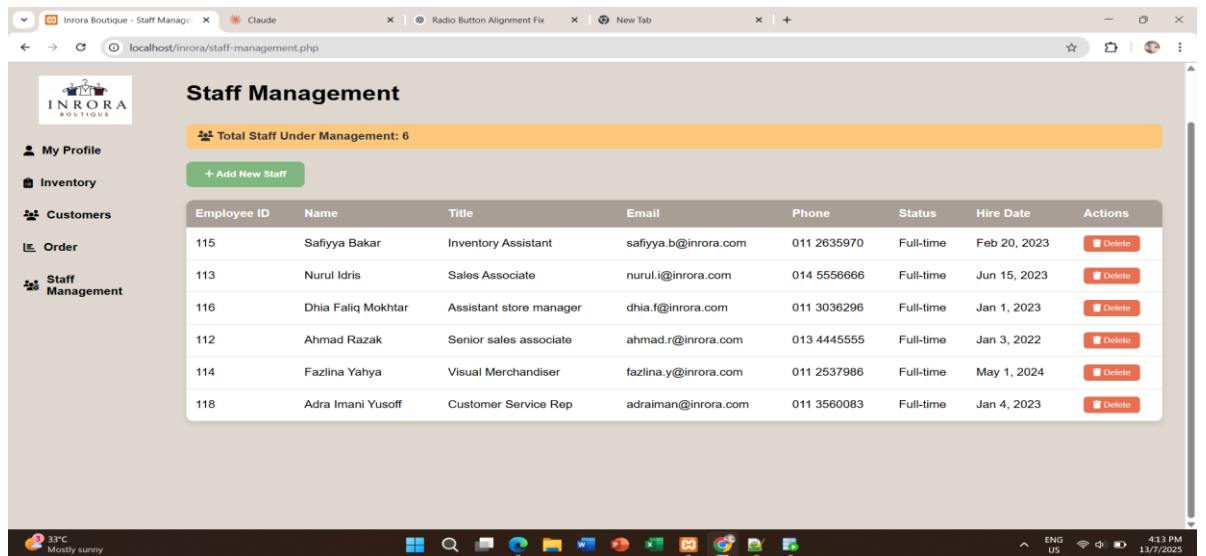


Figure 3.3.7 Delete Employee Page

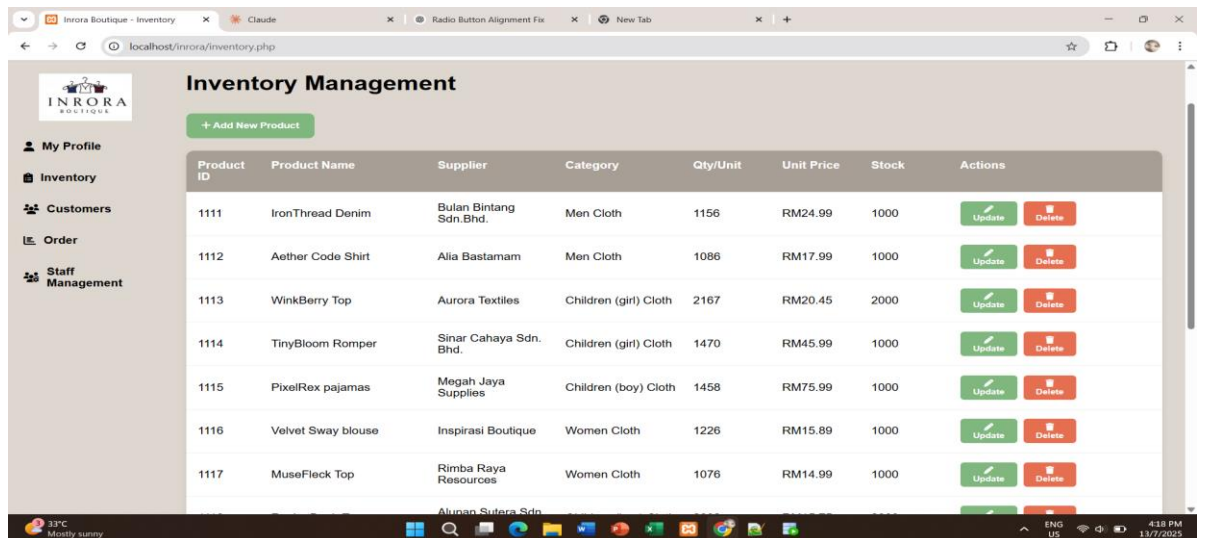


Figure 3.3.8 Read Product Page

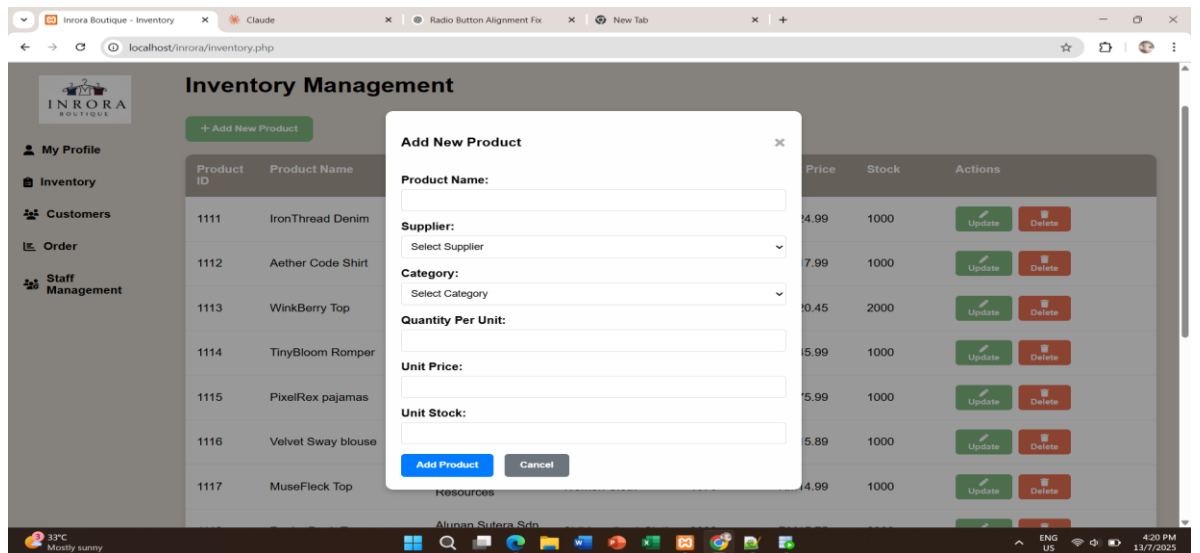


Figure 3.3.9 Insert Product Page

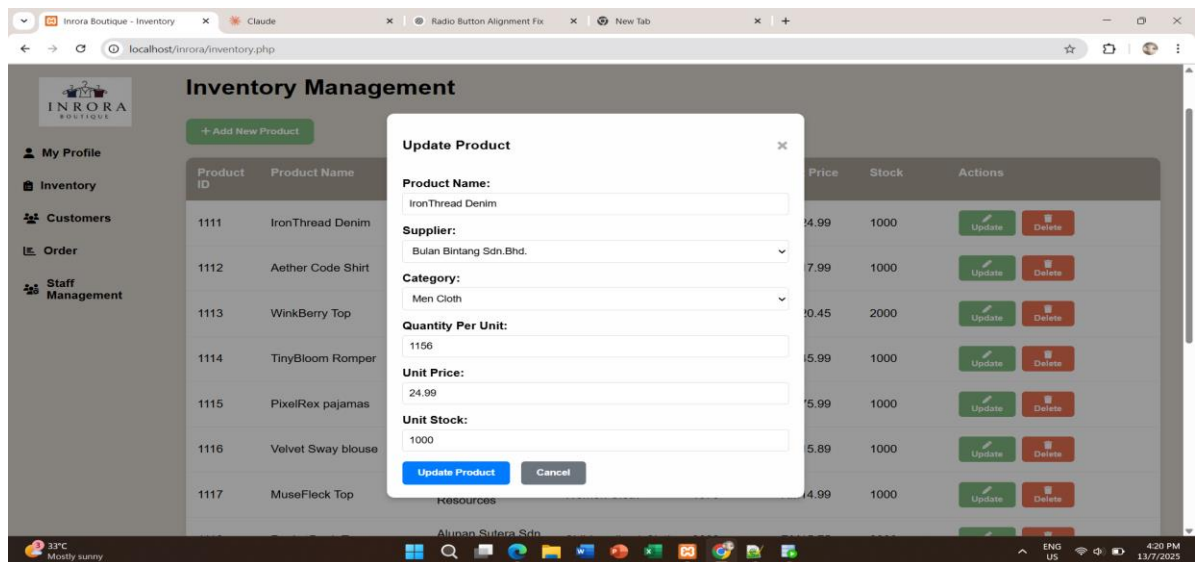


Figure 3.3.10 Update Product Page

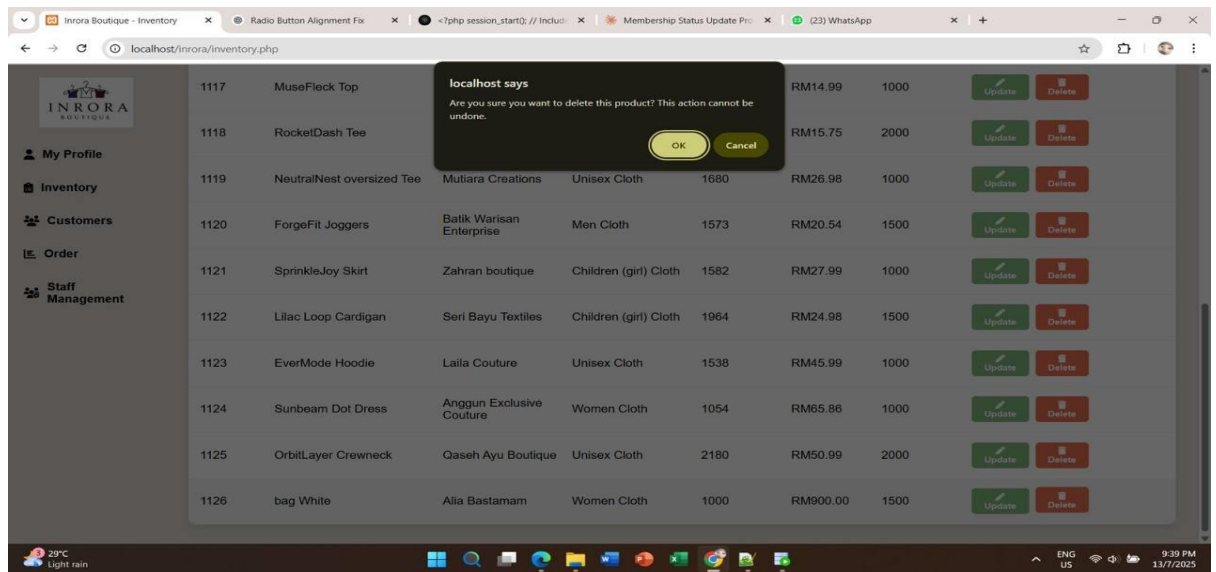


Figure 3.3.11 Delete Product Page (1)

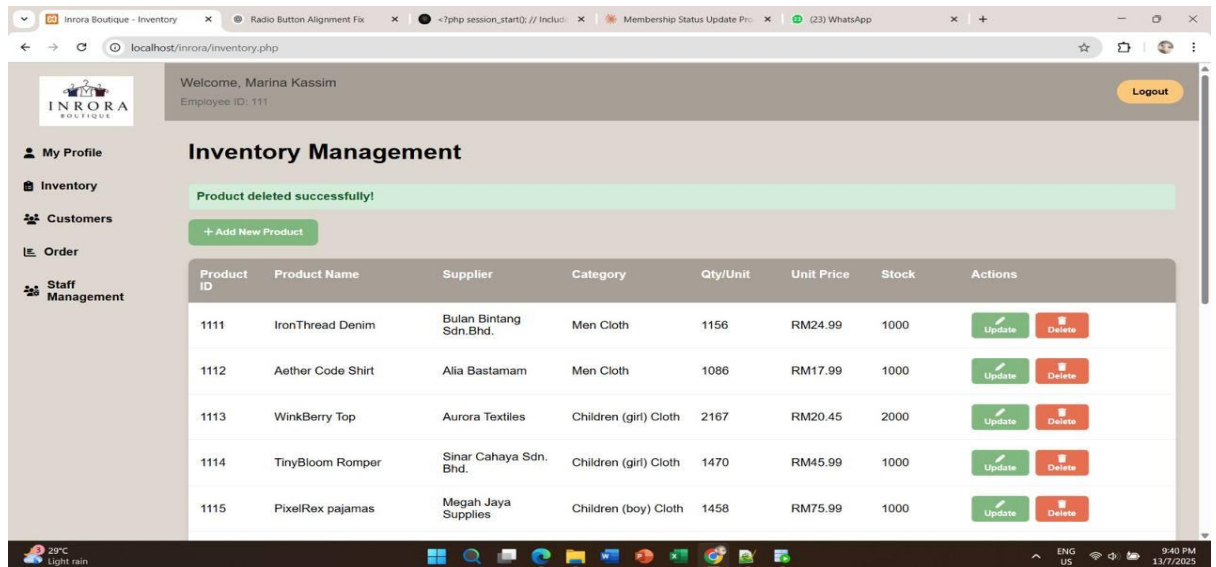


Figure 3.3.12 Delete Product Page (2)

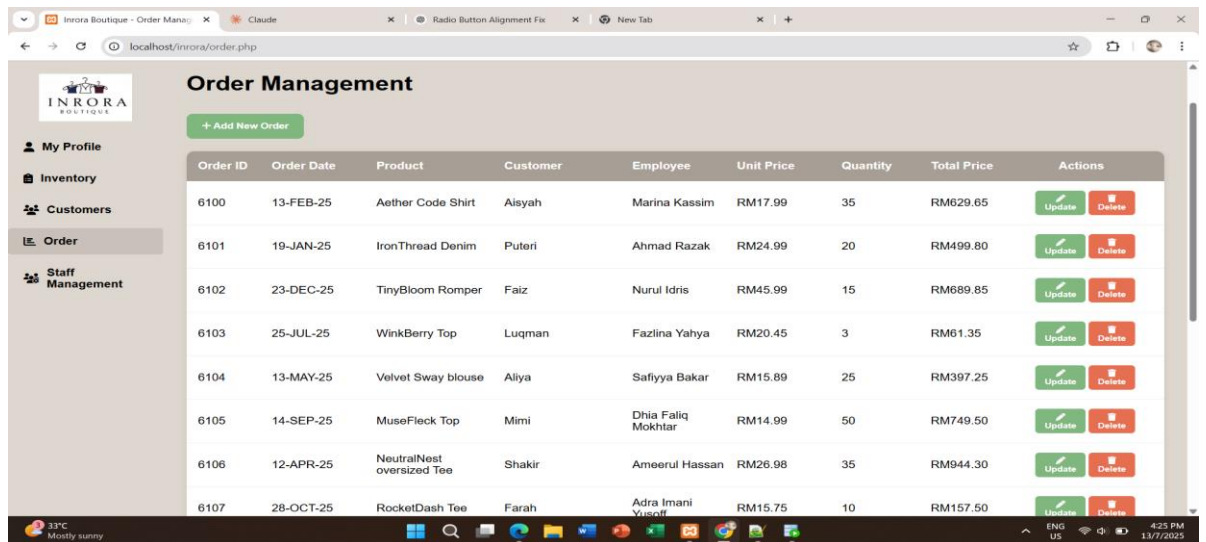


Figure 3.3.13 Read Order Page

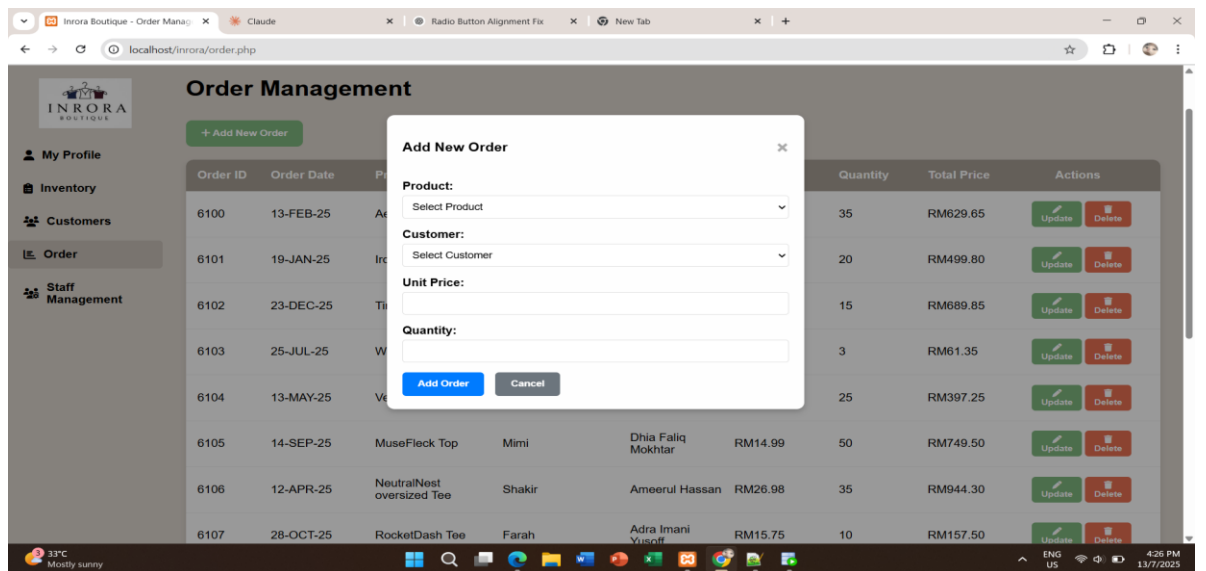


Figure 3.3.14 Insert Order Page

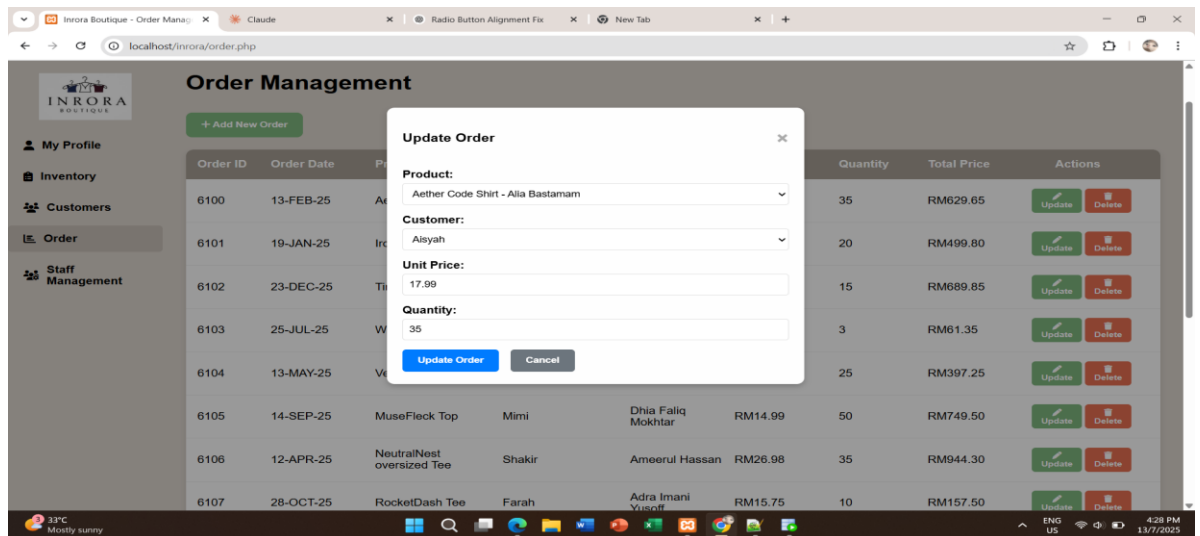


Figure 3.3.15 Update Order Page

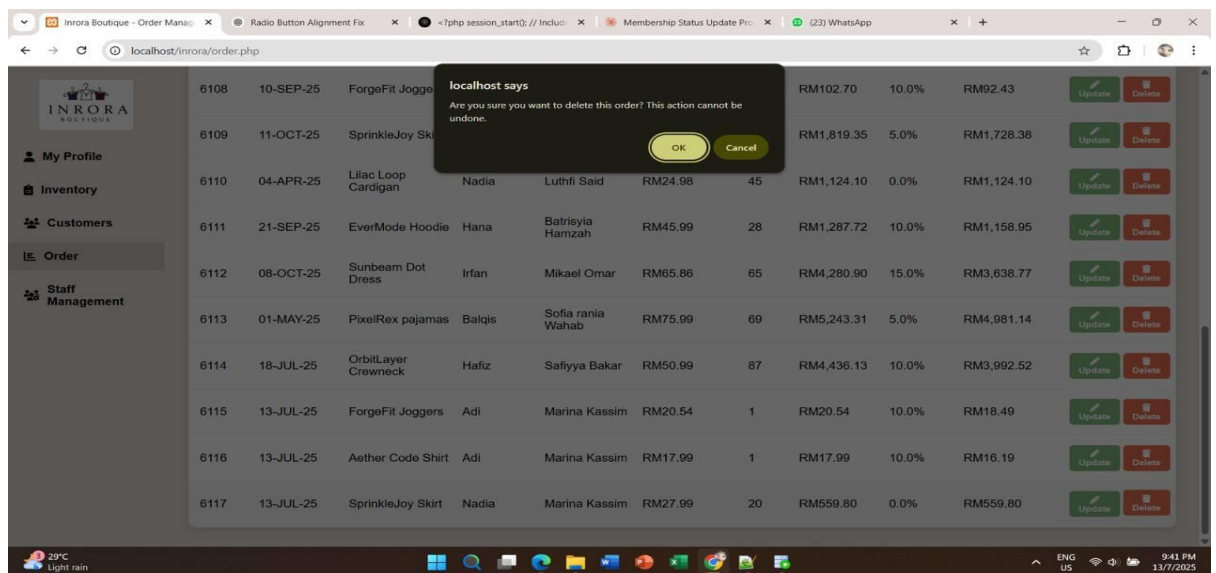


Figure 3.3.16 Delete Order Page (1)

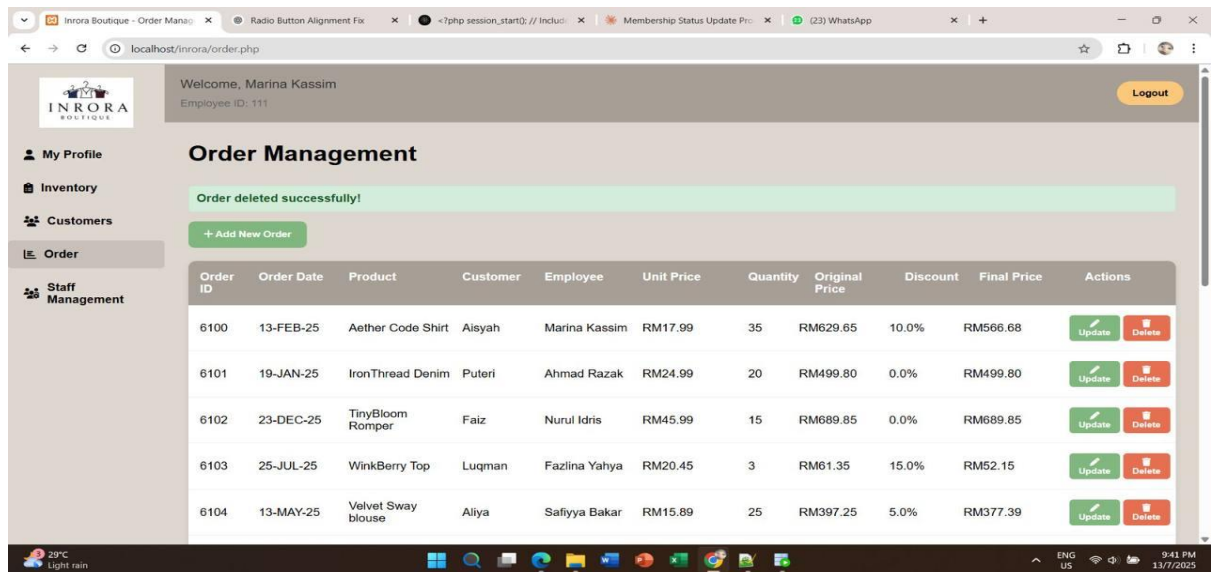


Figure 3.3.17 Delete Order Page (2)

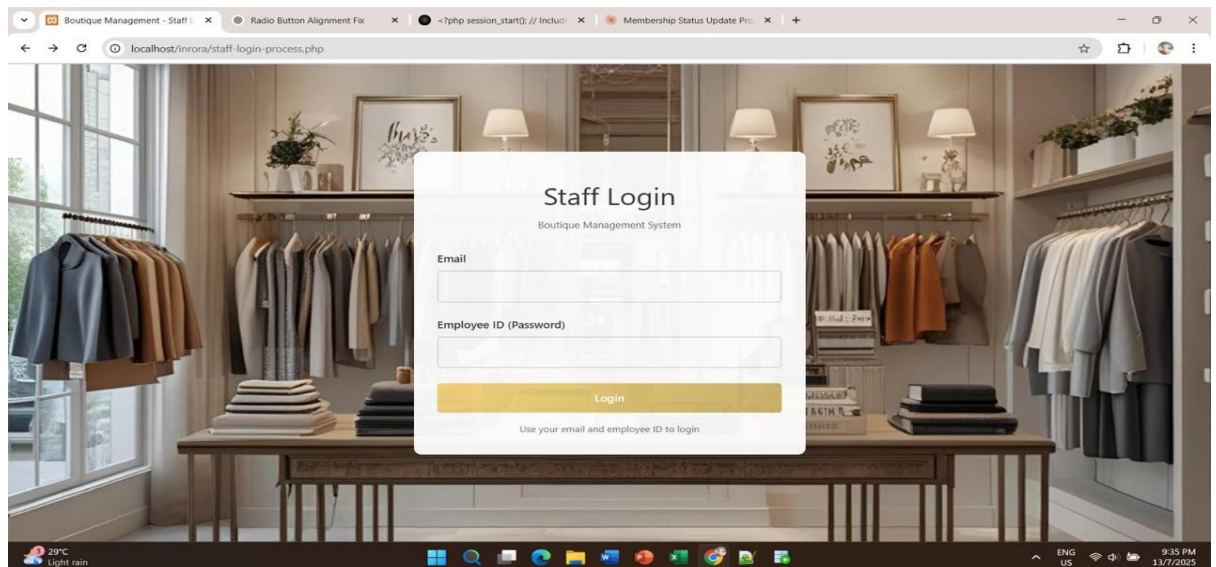


Figure 3.3.18 Staff's Login Page

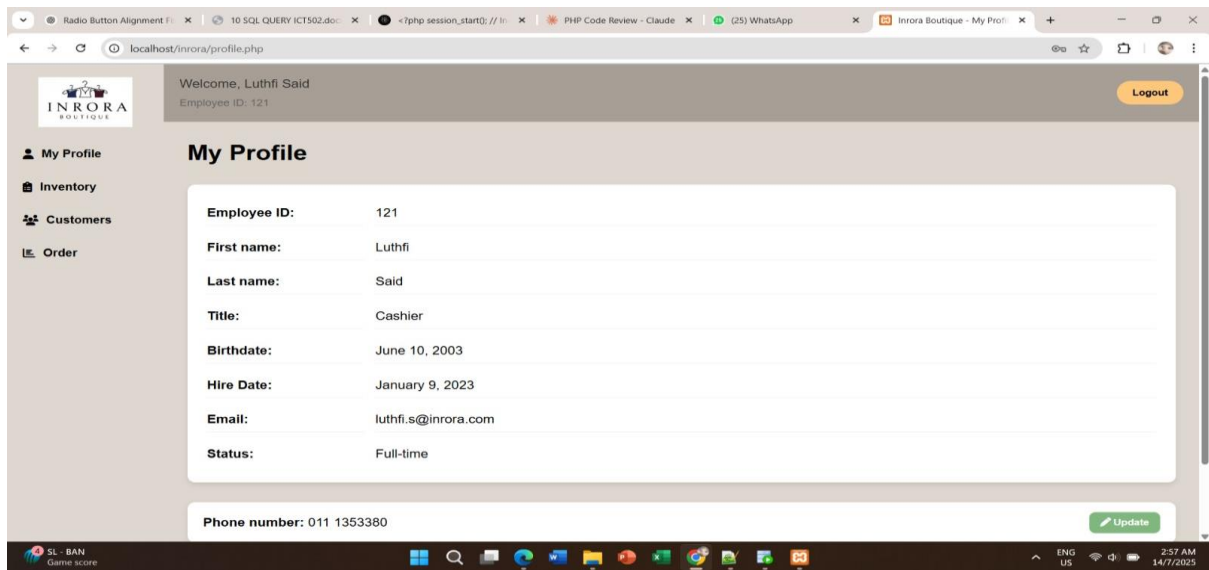


Figure 3.3.19 Recursive (1)

ManagerID creates a recursive relationship in the EMPLOYEES table. If an employee (like Luthfi) is not a manager, the Staff Management menu is hidden.

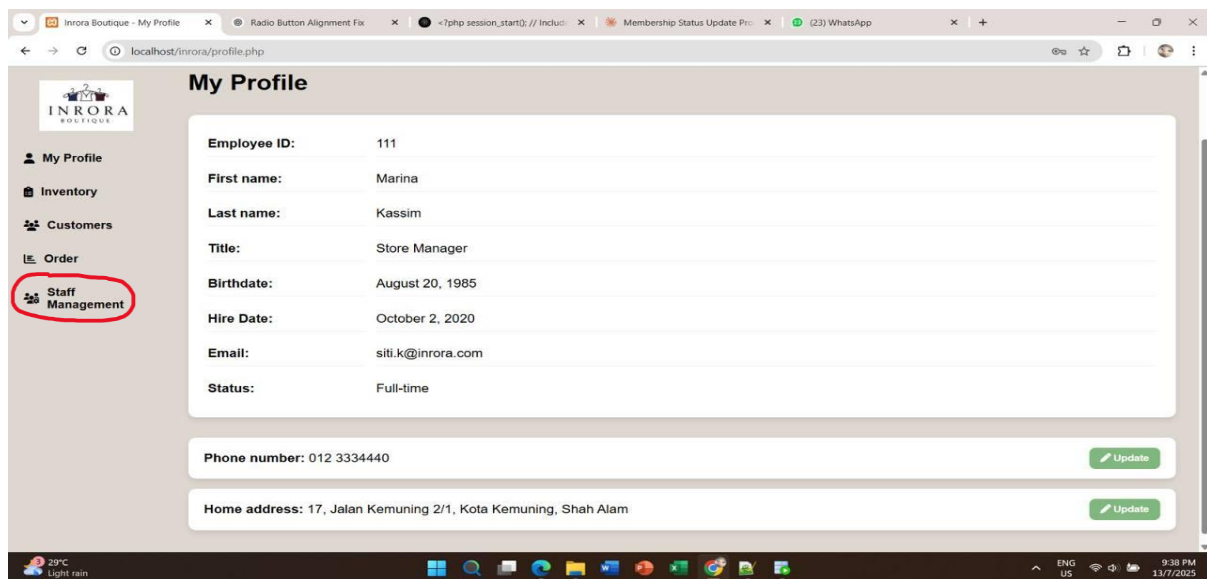


Figure 3.3.20 Recursive (2)

This figure shows that Marina is a manager which the Staff Management menu is visible.

The screenshot shows the 'Order Management' page of the Inrora Boutique system. It features a sidebar with navigation options: My Profile, Inventory, Customers, Order (selected), and Staff Management. The main area displays a table of orders with the following columns: Order ID, Order Date, Product, Customer, Employee, Unit Price, Quantity, Original Price, Discount, Final Price, and Actions. The table contains 10 rows of order data, each with 'Update' and 'Delete' buttons in the Actions column.

Order ID	Order Date	Product	Customer	Employee	Unit Price	Quantity	Original Price	Discount	Final Price	Actions
6100	13-FEB-25	Aether Code Shirt	Aisyah	Marina Kassim	RM17.99	35	RM629.65	10.0%	RM566.68	[Update] [Delete]
6101	19-JAN-25	IronThread Denim	Puteri	Ahmad Razak	RM24.99	20	RM499.80	0.0%	RM499.80	[Update] [Delete]
6102	23-DEC-25	TinyBloom Romper	Faiz	Nurul Idris	RM45.99	15	RM689.85	0.0%	RM689.85	[Update] [Delete]
6103	25-JUL-25	WinkBerry Top	Luqman	Fazlina Yahya	RM20.45	3	RM61.35	15.0%	RM52.15	[Update] [Delete]
6104	13-MAY-25	Velvet Sway blouse	Aliya	Safiyya Bakar	RM15.89	25	RM397.25	5.0%	RM377.39	[Update] [Delete]
6105	14-SEP-25	MuseFleck Top	Mimi	Dhia Faliq Mokhtar	RM14.99	50	RM749.50	0.0%	RM749.50	[Update] [Delete]
6106	12-APR-25	NeutralNest oversized Tee	Shakir	Ameerul Hassan	RM26.98	35	RM944.30	0.0%	RM944.30	[Update] [Delete]
6107	28-OCT-25	RocketDash Tee	Farah	Adra Imani	RM15.75	10	RM157.50	20.0%	RM126.00	[Update] [Delete]

Figure 3.3.21 Bridge

The bridge is order_details where it connects order table and products table as shown in this figure. The relation between these table allows for a customer to order multiple products while also allowing product can appear in many different orders. A many to many relationships can be identified from the explained relation due to the usage of bridge table, “order_details”.

The screenshot shows the 'Customer Management' page of the Inrora Boutique system. It features a sidebar with navigation options: My Profile, Inventory, Customers (selected), Order, and Staff Management. The main area displays a form for adding a customer. The form includes fields for Customer Name, Phone Number, Address, Email, and Contact Name. There are also radio buttons for Membership Options (No Membership, Add Membership) and a dropdown for Membership Discount (%). The form has 'Add Customer' and 'Cancel' buttons at the bottom.

Figure 3.3.22 Inheritance

The concept of inheritance can be seen where the customer is the parent table (superclass) and membership are the child table (subclass). Customer table stores the information for all customers and membership table stores only for customers who are members where store how many discounts they get.

Throughout the development process of the Project Management System, we have implemented various other database functions such as the usage of sequences to help us in generate a unique orderID for inserting new records into order table. An example of this implementation can be seen based on the Figure below taken from the source code of the system:

```
// First try to use sequence (recommended approach)
$seq_sql = "SELECT order_seq.NEXTVAL as next_id FROM DUAL";
$seq_stmt = oci_parse($dbconn, $seq_sql);

if (oci_execute($seq_stmt)) {
    $seq_result = oci_fetch_assoc($seq_stmt);
    $next_orderid = $seq_result['NEXT_ID'];
    oci_free_statement($seq_stmt);
} else {
    // If sequence doesn't exist, use MAX+1 approach
    oci_free_statement($seq_stmt);

    $max_sql = "SELECT NVL(MAX(ORDERID), 0) + 1 as next_id FROM ORDERS";
    $max_stmt = oci_parse($dbconn, $max_sql);
    oci_execute($max_stmt);
    $max_result = oci_fetch_assoc($max_stmt);
    $next_orderid = $max_result['NEXT_ID'];
    oci_free_statement($max_stmt);
}
```

Figure 3.3.23 Extra

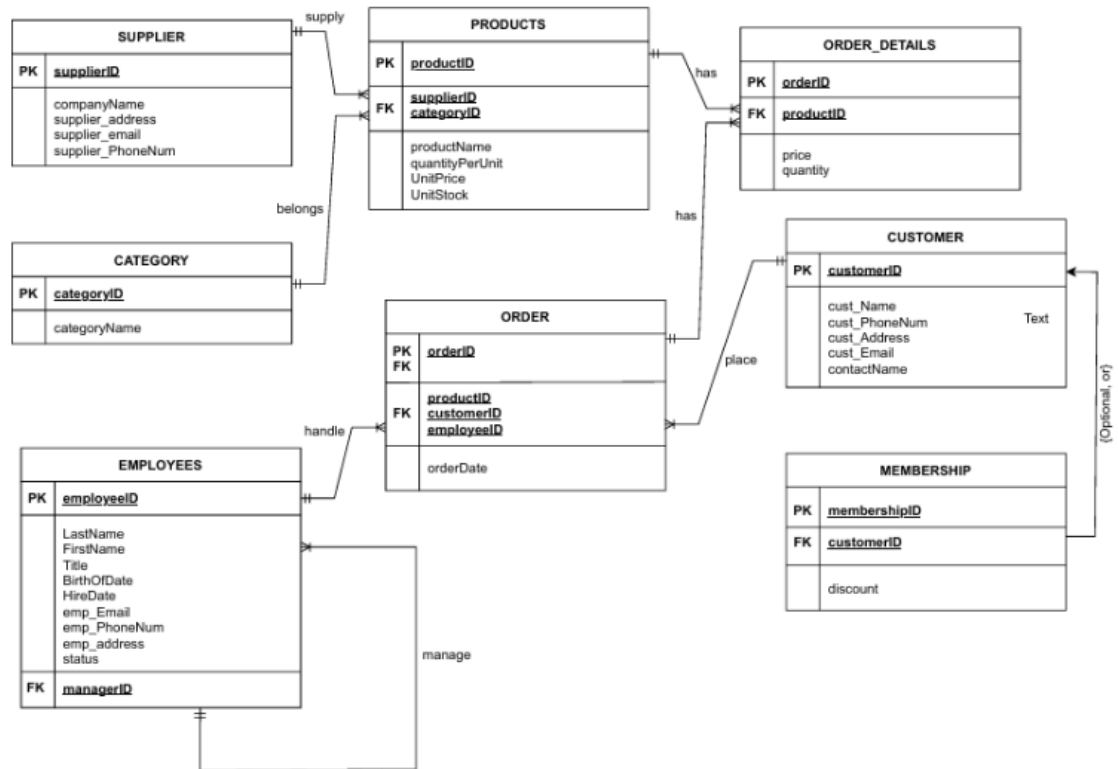
Based on the figure above, we have used the “.nextval” database function for sequence that have been created which is “order_seq.nextval as next_id” in our code to gets the next value from the order_seq sequence. If the sequence exists and the query executes successfully, the result(next value from the sequence) is fetched and assigned to \$next_orderid. Then, if the sequence does not exist or execution fails, the code falls back to an alternative method.

4.0 CONCLUSION

To sum up this report, we used Hypertext Markup Language (HTML) and Hypertext Preprocessor (PHP) to create a web-based system to develop an effective management system for Inrora Boutique. We have successfully identified the requirements that our system needs. Many of the factors were considered in designing the system such as minimalist theme, welcoming style and user-friendly. Generally, the system is functional and reduce the workload of the staff in Inrora Boutique. We have reconstructed a centralized and automated based system to replace manual tracking that were prone to error, slow and unscalable which perhaps will improve accurate the stock tracking in a time. The system has one types of users only which are employee of Inrora Boutique. The system checks out some time consuming on sales such as checkout the sales in manual led to price or calculation errors. The system also avoids inefficient membership and loyalty tracking where it will automatically apply discounts based on status of customers. The developed system perfectly met all the objectives conceived at the design phase of the system. Hence, we have completed our project without any of problem.

5.0 APPENDIX

5.1 Appendix A : ERD



5.2 Appendix B : Data Dictionary

CATEGORY

	↕ COLUMN_NAME	↕ DATA_TYPE	↕ NULLABLE	DATA_DEFAULT	↕ COLUMN_ID	↕ COMMENTS
1	CATEGORYID	NUMBER(38,0)	No	(null)	1 (null)	
2	CATEGORYNAME	VARCHAR2(100 BYTE)	Yes	(null)	2 (null)	

CUSTOMER

	↕ COLUMN_NAME	↕ DATA_TYPE	↕ NULLABLE	DATA_DEFAULT	↕ COLUMN_ID	↕ COMMENTS
1	CUSTOMERID	NUMBER(38,0)	No	(null)	1 (null)	
2	CUST_NAME	VARCHAR2(100 BYTE)	No	(null)	2 (null)	
3	CUST_PHONENUM	VARCHAR2(100 BYTE)	Yes	(null)	3 (null)	
4	CUST_ADDRESS	VARCHAR2(100 BYTE)	Yes	(null)	4 (null)	
5	CUST_EMAIL	VARCHAR2(100 BYTE)	Yes	(null)	5 (null)	
6	CONTACTNAME	VARCHAR2(100 BYTE)	Yes	(null)	6 (null)	

EMPLOYEES

	↕ COLUMN_NAME	↕ DATA_TYPE	↕ NULLABLE	DATA_DEFAULT	↕ COLUMN_ID	↕ COMMENTS
1	EMPLOYEEID	NUMBER(38,0)	No	(null)	1 (null)	
2	LASTNAME	VARCHAR2(100 BYTE)	No	(null)	2 (null)	
3	FIRSTNAME	VARCHAR2(100 BYTE)	Yes	(null)	3 (null)	
4	TITLE	VARCHAR2(100 BYTE)	Yes	(null)	4 (null)	
5	BIRTHOFDATE	DATE	Yes	(null)	5 (null)	
6	HIREDATE	DATE	Yes	(null)	6 (null)	
7	EMP_EMAIL	VARCHAR2(100 BYTE)	Yes	(null)	7 (null)	
8	EMP_PHONENUM	VARCHAR2(100 BYTE)	Yes	(null)	8 (null)	
9	EMP_ADDRESS	VARCHAR2(100 BYTE)	Yes	(null)	9 (null)	
10	STATUS	VARCHAR2(100 BYTE)	Yes	(null)	10 (null)	
11	MANAGERID	NUMBER(38,0)	Yes	(null)	11 (null)	

MEMBERSHIP

	↕ COLUMN_NAME	↕ DATA_TYPE	↕ NULLABLE	DATA_DEFAULT	↕ COLUMN_ID	↕ COMMENTS
1	MEMBERSHIPID	NUMBER(38,0)	No	(null)	1 (null)	
2	CUSTOMERID	NUMBER(38,0)	No	(null)	2 (null)	
3	DISCOUNT	BINARY_FLOAT	Yes	(null)	3 (null)	

ORDER_DETAILS

	↕ COLUMN_NAME	↕ DATA_TYPE	↕ NULLABLE	DATA_DEFAULT	↕ COLUMN_ID	↕ COMMENTS
1	ORDERID	NUMBER(38,0)	No	(null)	1	(null)
2	PRODUCTID	NUMBER(38,0)	No	(null)	2	(null)
3	PRICE	BINARY_FLOAT	Yes	(null)	3	(null)
4	QUANTITY	NUMBER(38,0)	Yes	(null)	4	(null)

ORDERS

	↕ COLUMN_NAME	↕ DATA_TYPE	↕ NULLABLE	DATA_DEFAULT	↕ COLUMN_ID	↕ COMMENTS
1	ORDERID	NUMBER(38,0)	No	(null)	1	(null)
2	ORDERDATE	DATE	No	(null)	2	(null)
3	PRODUCTID	NUMBER(38,0)	Yes	(null)	3	(null)
4	CUSTOMERID	NUMBER(38,0)	Yes	(null)	4	(null)
5	EMPLOYEEID	NUMBER(38,0)	Yes	(null)	5	(null)

PRODUCTS

	↕ COLUMN_NAME	↕ DATA_TYPE	↕ NULLABLE	DATA_DEFAULT	↕ COLUMN_ID	↕ COMMENTS
1	PRODUCTID	NUMBER(38,0)	No	(null)	1	(null)
2	PRODUCTNAME	VARCHAR2(100 BYTE)	No	(null)	2	(null)
3	QUANTITYPERUNIT	NUMBER(38,0)	Yes	(null)	3	(null)
4	UNITPRICE	BINARY_FLOAT	Yes	(null)	4	(null)
5	SUPPLIERID	NUMBER(38,0)	Yes	(null)	5	(null)
6	CATEGORYID	NUMBER(38,0)	Yes	(null)	6	(null)
7	UNITSTOCK	NUMBER(38,0)	Yes	(null)	7	(null)

SUPPLIER

	↕ COLUMN_NAME	↕ DATA_TYPE	↕ NULLABLE	DATA_DEFAULT	↕ COLUMN_ID	↕ COMMENTS
1	SUPPLIERID	NUMBER(38,0)	No	(null)	1	(null)
2	COMPANYNAME	VARCHAR2(100 BYTE)	No	(null)	2	(null)
3	SUPPLIER_ADDRESS	VARCHAR2(100 BYTE)	Yes	(null)	3	(null)
4	SUPPLIER_EMAIL	VARCHAR2(100 BYTE)	Yes	(null)	4	(null)
5	SUPPLIER_PHONENUM	VARCHAR2(100 BYTE)	Yes	(null)	5	(null)

5.3 Appendix C : DDL

1. CREATE, DROP table CATEGORY

```
CREATE TABLE CATEGORY (  
  categoryID NUMBER(38),  
  category_name VARCHAR(100),  
  CONSTRAINT categoryID_pk PRIMARY KEY(categoryID)  
);  
  
DROP TABLE CATEGORY;
```

2. CREATE, DROP table CUSTOMER

```
CREATE TABLE CUSTOMER (  
  customerID NUMBER(38),  
  cust_Name VARCHAR (100),  
  cust_PhoneNum VARCHAR(100),  
  cust_address VARCHAR(100),  
  cust_Email VARCHAR(100),  
  contactName VARCHAR(100),  
  CONSTRAINT customerID_pk PRIMARY KEY (customerID)  
);  
  
DROP TABLE CUSTOMER;
```

3. CREATE, DROP table MEMBERSHIP

```
CREATE TABLE MEMBERSHIP (  
  membershipID NUMBER(38),  
  customerID NUMBER(38),  
  discount BINARY_FLOAT,  
  CONSTRAINT membershipID_pk PRIMARY KEY (membershipID),  
  CONSTRAINT customerID_fk FOREIGN KEY (customerID) REFERENCES customer(customerID)  
);  
  
DROP TABLE MEMBERSHIP;
```

4. CREATE, DROP table EMPLOYEES

```
CREATE TABLE EMPLOYEES (  
  employeeID NUMBER(38),  
  LastName VARCHAR(100),  
  Firstname VARCHAR(100),  
  Title VARCHAR(100),  
  BirtOfDate DATE,  
  HireDate DATE,  
  emp_Email VARCHAR(100),  
  emp_PhoneNum VARCHAR(100),  
  emp_address VARCHAR(100),  
  status VARCHAR (100),  
  managerID NUMBER(38),  
  CONSTRAINT employeeID_pk PRIMARY KEY (employeeID),  
  CONSTRAINT managerID_fk FOREIGN KEY(managerID) REFERENCES employees(employeeID));  
  
DROP TABLE EMPLOYEES;
```

5. CREATE,DROP table ORDER_DETAILS

```
CREATE TABLE ORDER_DETAILS(  
  Orderid NUMBER (38),  
  Productid NUMBER(38),  
  Price BINARY_FLOAT,  
  Quantity NUMBER (38),  
  CONSTRAINT orderid_pk PRIMARY KEY (orderid),  
  CONSTRAINT productid_fk FOREIGN KEY(productid) REFERENCES PRODUCTS(productid));  
  
DROP TABLE ORDER_DETAILS;
```

6. CREATE, DROP table ORDERS

```
CREATE TABLE ORDERS(  
  Orderid number(38),  
  Orderdate date default SYSDATE,  
  Productid number(38),  
  Customerid number(38),  
  Employeeid number(38),  
  CONSTRAINT orderid_pk PRIMARY KEY (orderid),  
  CONSTRAINT customerid_fk FOREIGN KEY(customerid) REFERENCES CUSTOMER (customerid),  
  CONSTRAINT employeeid_fk FOREIGN KEY (employeeid) REFERENCES EMPLOYEES (employeeid) );  
  
DROP TABLE ORDERS;
```


7. CREATE, DROP table PRODUCTS

```
CREATE TABLE PRODUCTS(  
    Productid number(38),  
    Supplierid number(38),  
    categoryid number(38),  
    Productname varchar(100),  
    Quantityperunit number(38),  
    Unit_price binary_float,  
    Unitstock number(38),  
    CONSTRAINT productid_pk PRIMARY KEY (productid),  
    CONSTRAINT categoryid_fk FOREIGN KEY (categoryid) REFERENCES CATEGORY (categoryid),  
    CONSTRAINT productname_nn CHECK (productname IS NOT NULL) );  
  
DROP TABLE PRODUCTS;
```

8. CREATE, DROP table SUPPLIER

```
CREATE TABLE SUPPLIER (  
    Supplierid number(38),  
    Companyname varchar2 (100),  
    Supplier_address varchar2(100),  
    Supplier_email varchar2(100),  
    Supplier_phonenum varchar2(100),  
    CONSTRAINT supplierid_pk PRIMARY KEY (supplierid) );  
  
DROP TABLE SUPPLIER;
```


5.4 Appendix D : DML

1. SELECT, INSERT, DELETE table CATEGORY

```
SELECT* FROM CATEGORY WHERE categoryName LIKE 'Children%';
```

```
DELETE FROM CATEGORY;
```

2. SELECT, INSERT, UPDATE, DELETE table CUSTOMER

```
SELECT* FROM CUSTOMER WHERE cust_Address LIKE '%Perak';
```

```
INSERT INTO CUSTOMER (customerID, cust_Name, cust_PhoneNNum,  
cust_Address, cust_Email, contactName)  
VALUES (216, 'Jazlina', '0109604545', 'No.38, Jalan Mawar, 96000 Sibul, Sarawak',  
'jazlina@gmail.com', 'Miss Jas');
```

```
UPDATE CUSTOMER  
SET cust_PhoneNum = '0182738294'  
WHERE customerID = 211;
```

```
DELETE FROM CUSTOMER;
```

3. SELECT, INSERT, UPDATE, DELETE table EMPLOYEES

```
SELECT* FROM EMPLOYEES WHERE TO_CHAR(HireDate,'MM') = 1;
```

```
INSERT INTO EMPLOYEES (employeeID, LastName, FirstName, Title,  
BirthOfDate, HireDate, emp_email, emp_PhoneNum, emp_address, status,  
managerID)  
VALUES (126, 'Mahmud', 'Jamal', 'Junior Sales Associate', 25/7/2006,  
'mahmud@gmail.com', '0198393004', '36B Lorong Sultan, 50450 Ampang, Kuala  
Lumpur', 'Full-Time', 123);
```

```
UPDATE EMPLOYEES SET Title = 'Assistant Manager' WHERE employeeID =  
112;
```

```
DELETE FROM EMPLOYEES;
```

4. SELECT, INSERT, UPDATE, DELETE table MEMBERSHIP

```
SELECT* FROM MEMBERSHIP WHERE customerID IN(201,203,205,207,209);
```

```
INSERT INTO MEMBERSHIP (membershipID, customerID, discount)  
VALUES (511, 210, 0.1);
```

```
UPDATE MEMBERSHIP SET discount = 0.1 WHERE membershipID = 503;
```

```
DELETE FROM MEMBERSHIP;
```

5. SELECT, INSERT, UPDATE, DELETE table ORDERS

```
SELECT* FROM ORDERS WHERE TO_CHAR(orderDate, 'MM') = 7;
```

```
INSERT INTO ORDERS (orderID, productID, customerID, employeeID, orderDate)  
VALUES (6115, 1112, 212, 112, 3/4/2025);
```

```
UPDATE ORDERS SET productID = 1114 WHERE orderID = 6106;
```

```
DELETE FROM ORDERS;
```

6. SELECT, INSERT, UPDATE, DELETE table ORDER_DETAILS

```
SELECT* FROM ORDER_DETAILS WHERE quantity > 40;
```

```
INSERT INTO ORDER_DETAILS (orderID, productID, price, quantity)  
VALUES (6115, 1112, 17.99, 15)
```

```
UPDATE ORDER SET quantity = 20 WHERE orderID = 6106;
```

```
DELETE FROM ORDER_DETAILS;
```

7. SELECT, INSERT, UPDATE, DELETE from PRODUCTS

```
SELECT* FROM PRODUCTS WHERE productName LIKE '___n%';
```

```
INSERT INTO PRODUCTS (productID, supplierID, categoryID, productName,  
quantityPerUnit, unitPrice, unitStock)  
VALUES (1126, 16, 813, 'Twinkle Tots', 1500, 25.50, 1000);
```

```
UPDATE PRODUCTS SET unitPrice = 21.99 WHERE productID = 1112;
```

DELETE FROM PRODUCTS;

8. SELECT from SUPPLIER

SELECT* FROM SUPPLIER WHERE address LIKE '%Kuala Lumpur';